Python Assignments and Exercises Report

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Exercises 1-41: Fundamentals, Files, OOP, Data, Flask, and MQTT
Repository: https://github.com/chittarad06-cell/ERICO-Exercises

Abstract

This report summarizes the completed Python assignment exercises. The exercises are grouped by topic so the repository is easy to review and the learning progression is clear.

Exercises	File or folder	Implementation strength
1-4	01_python_basics.py	functions, recursion, lambda, lists, tuples, range, break, continue
5-6	02_files_regex_encoding.py	regex, pathlib, Base64, ASCII, URL encoding, ROT13, CSV, JSON, binary files
7-12	03_oop_logging_collections.py	dataclass, inheritance, logging, deque, custom exceptions
13-21	04_data_analysis.py	NumPy, pandas, filtering, groupby, pivot table, normalization, rolling average
22-30	flask_portfolio_app	routes, templates, forms, sessions, JSON POST validation, Docker/Procfile
31-41	mqtt_examples	MQTT report, topic design, publisher/subscriber, callbacks, QoS, threading

Exercises 1-4: Python Fundamentals

- variables and type conversion
- functions, recursion, lambda
- range-based repetition and filtering

These exercises establish the foundation: how to store values, transform values, reuse logic, and control program flow.

Exercises 5-6: Files, Regex, and Encoding

- regex for pattern extraction
- pathlib for file handling
- Base64/ASCII/URL/ROT13 round trips
- CSV, JSON, and binary file examples

This section shows Python as an automation tool for real text and file processing tasks.

Exercises 7-12: OOP and Reliability

- inheritance and method overriding
- encapsulation and class methods
- logging levels
- stack, queue, deque, custom exceptions

This section connects Python syntax with software quality through classes, logging, collections, and safe error handling.

Exercises 13-21: Data Analysis

- NumPy statistics
- pandas cleaning and grouping
- pivot table and normalization
- time-series rolling average

This section shows why data must be cleaned, transformed, summarized, and checked before conclusions are drawn.

Exercises 22-30: Flask

- routes, query strings, templates
- forms and sessions
- JSON POST validation
- Docker and deployment files

This section demonstrates how Python can serve web pages, accept user input, validate API data, and prepare for deployment.

Exercises 31-41: MQTT

- MQTT vs HTTP comparison
- smart-home topic hierarchy
- publisher/subscriber callbacks
- QoS, security, and threading notes

This section introduces event-driven IoT communication and shows why MQTT fits repeated sensor messages better than normal request-response HTTP.

Conclusion

The exercises show a complete path from basic Python to practical automation, data analysis, web applications, and IoT communication. The work emphasizes choosing the correct construct for each task.

Submission Evidence

- Repository includes code for exercises 1-41 from the visible assignment sheet.
- README explains topic grouping and why specific constructs were used.
- This report documents implementation choices and learning outcomes.
- VIDEO_GUIDE.md is included to support the short explanation video.